

PoE Speaker Rev E

Final Manufacturing & Build Notes

Document	Manufacturing / Assembly Notes
Design	PoE-Speaker-RevE
PCB CAD	KiCad
Release	Final Manufacturing Package
PCB Type	4-Layer Controlled-Impedance PCB
Fabrication Target	JLCPCB / Compatible PCB Manufacturer

1. PCB Fabrication

The PCB shall be manufactured using the final fabrication data contained in this release package.

- **Layer count:** 4 layers
- **Final controlled-impedance stack-up:** JLC04161H-7628A
- **Ethernet controlled impedance:** 100 Ω differential
- **Ethernet differential trace width:** 0.2764 mm
- **Ethernet differential pair spacing:** 0.20 mm
- **Outer copper:** 1 oz
- **Inner copper:** 0.5 oz

The Gerber and drill files supplied in this package shall be used as the definitive PCB fabrication outputs.

2. Controlled-Impedance Stack-Up

The final PCB stack-up is based on the **JLCPCB JLC04161H-7628A** controlled-impedance construction.

Layer / Material	Type	Thickness
F.Cu	Copper	0.0350 mm
FR4 7628	Prepreg	0.2180 mm
FR4 1080	Prepreg	0.0764 mm
In1.Cu	Copper	0.0152 mm
FR4	Core	0.8650 mm
In2.Cu	Copper	0.0152 mm
FR4 1080	Prepreg	0.0764 mm
FR4 7628	Prepreg	0.2180 mm
B.Cu	Copper	0.0350 mm

CONTROLLED-IMPEDANCE REQUIREMENT

The Ethernet differential pairs are designed for a nominal **100 Ω differential impedance**.
The final routing geometry is:

- Trace width: **0.2764 mm**
- Pair spacing: **0.20 mm**
- Signal layer: **L1 / F.Cu**
- Reference layer: **L2 / In1.Cu**

This geometry is associated with the **JLC04161H-7628A** stack-up specified above.

3. Assembly Data

Use the assembly data included in this final manufacturing package for component sourcing and placement.

- Use the supplied **Bill of Materials (BOM)** for component identification and sourcing.
- Manufacturer part numbers contained in the BOM shall be used where specified.
- Use the supplied **CPL / Pick-and-Place file** for automated component placement.
- Component reference designators and orientations shall follow the final KiCad PCB project.
- The supplied **STEP model** may be used for mechanical integration and assembly reference.

4. Amplifier Thermal Requirement

The audio power amplifier requires appropriate thermal management in the final assembly.

Reference	U5
Device	TPA3116D2DADR
Thermal Requirement	External top-side heatsink required in final assembly

The final mechanical assembly shall provide suitable heatsink mounting and adequate thermal clearance around the amplifier.

5. Design Verification

The final PCB design was checked after completion of the routing and controlled-impedance stack-up configuration.

- Final PCB Design Rules Check (**DRC**) performed.
- Final DRC result: **0 violations**.
- Final DRC result: **0 unconnected items**.
- Final schematic Electrical Rules Check (**ERC**) performed before release.

Fresh DRC/ERC verification results are included with the final manufacturing package.

6. Power Routing

The final power distribution routing was reviewed before release.

High-current 24 V paths were routed using widened traces and/or large copper areas where required. The reviewed power distribution includes:

- 24V_POE
- 24V_BENCH_PROTECTED
- 24V_OR
- 24V_AMP
- 3.3 V distribution

Narrow high-current sections in the main amplifier supply path were reviewed during final PCB cleanup.

7. Ethernet Routing

The Ethernet interface routing was reviewed and updated before the manufacturing release. The Ethernet TX and RX differential pairs use the controlled geometry specified in this document. Both conductors of each differential pair shall maintain consistent routing geometry wherever practical. The final controlled geometry is:

Parameter	Final Value
Differential impedance	100 Ω
Trace width	0.2764 mm
Pair spacing	0.20 mm
Signal layer	F.Cu / L1
Reference layer	In1.Cu / L2

8. Manufacturing Package Contents

The final manufacturing release package contains the following deliverables:

- Final KiCad project and source files
- Gerber fabrication files
- PTH drill file
- NPTH drill file
- Gerber job file
- Bill of Materials (BOM)
- CPL / Pick-and-Place file
- STEP mechanical model
- Final DRC verification result
- Final ERC verification result
- Manufacturing and build notes

9. Production Release

IMPORTANT — SOURCE OF TRUTH

The KiCad project and manufacturing outputs contained in this final release package constitute the **definitive manufacturing version** of the PoE-Speaker-RevE PCB.

Earlier project ZIP files, Gerbers, drill files, BOM files, CPL files, STEP exports, or other manufacturing outputs shall **not** be used for production.

Project: PoE-Speaker-RevE

Release: Final Manufacturing Package

END OF MANUFACTURING / BUILD NOTES